



BITS Pilani

Automotive Vehicle

16-08-2026

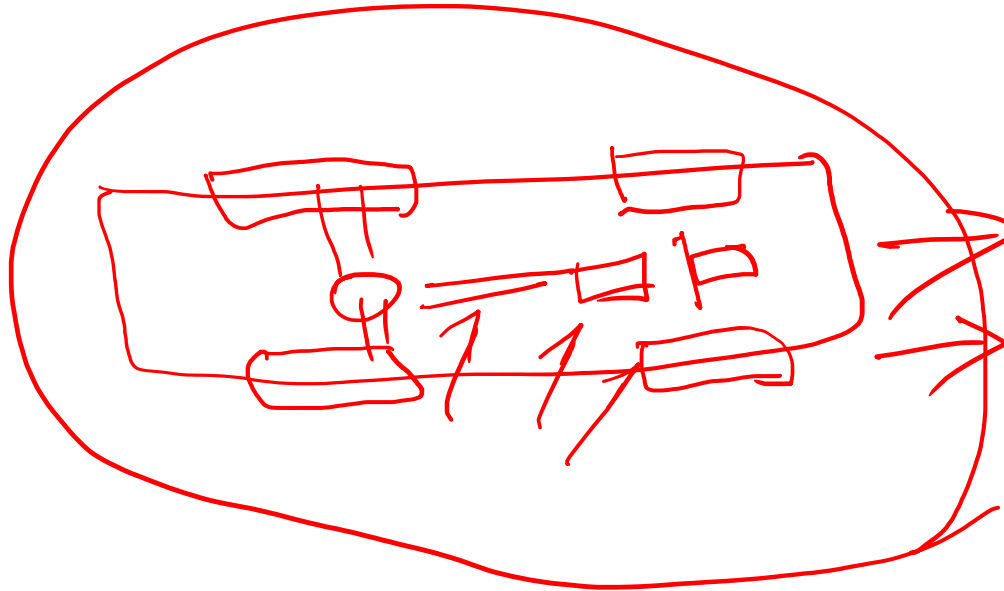
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Department of Mechanical Engineering



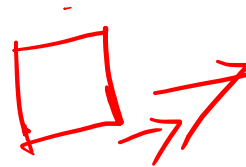
Automotive Vehicle [AELZC441]

Lecture No. 04: Introduction & Chassis-body

Recap of Previous session



→ Grade
 → Air
 → Rolling
 —



Handout summary

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics
1 <i>2hr</i>	Introduction to automobiles	- Automobiles, classification, and homologation - Development methodology, virtual engineering methods
2	Chassis and Body	- Types of chassis and applications - Weight distribution and analysis
3 <i>2hr</i>	Powertrain Fundamentals	- Vehicle propulsion system requirements - Tractive force and torque requirement - Powertrain architecture

4 hr
4, 5, 6
Enging ✓

Introduction

1. Automobile Classification

2. Homologation

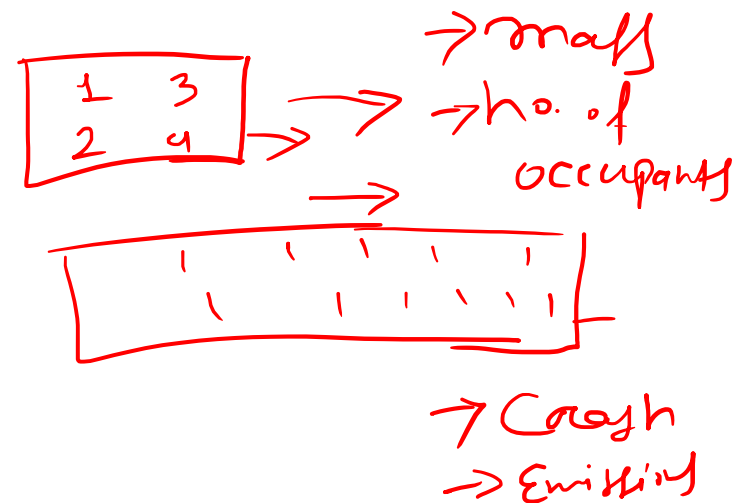
3. Development Methodology

4. Virtual Engineering Methods

Session Roadmap

1. Automobile Classification

Why?



Why Classification

➤ **Classification determines the rulebook**

It fixes which crash, emission, noise, and safety standards a vehicle must meet.

➤ **It determines cost structure**

Vehicle category drives taxation, duties, and registration fees in most markets.

➤ **It shapes engineering targets**

Category limits (mass, seating, power-to-weight) become hard constraints.

➤ **It is not globally uniform**

The same physical vehicle can fall into different regulatory categories in different markets.

Classification Types

Purpose & capacity ✓

Configuration ✓

Fuel source ✓

Body style ✓

Side of drive ✓

~~Based on capacity~~ ✓

Transmission type ✓

Regulatory criteria ✓

Emissions ✓



Classification: Based on Purpose & capacity

➤ Passenger Vehicles

Carry people

➤ Good Vehicles

Carry objects / goods

Classification	GVW
Light	$\leq 7,500$ kg
Medium	$>7,500$ to $\leq 12,000$ kg
Heavy	$>12,000$ kg

LMV →

Light motor vehicle

MOV →

Medium goods vehicles

HGV

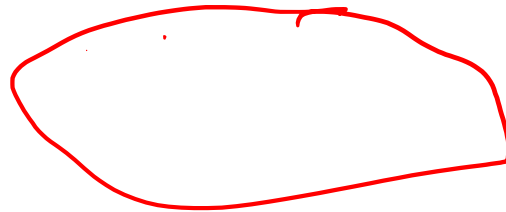
✓ GVW (Gross Vehicle Weight) = Maximum permissible total weight of the vehicle

= vehicle self weight + passengers +
trailer + Cargo / luggages

Source: Motor Vehicles Act, 1988 - official India Code PDF

Classification: Based on Configuration

- **Single-Unit Vehicle:** Engine and passenger/cargo space are part of one complete vehicle unit.



- **Articulated Vehicle:** Consists of **two or more connected sections**, with a joint allowing relative movement.



better turning



Classification: Based on Fuel source

A. Petrol/Diesel Engine Vehicles:

Powered by petrol/diesel

PM

Prime mover
↳ who is responsible to ~~propel~~ provide power

B. Gas Turbine Vehicles:

Vehicles that use a gas turbine as the power source.



C. Solar Vehicles:

Vehicles that are powered by solar energy.



↳ Vayhe

Classification: Based on Fuel source

D. Hydrogen Vehicles:

Fuel cell PEM

Vehicles that use hydrogen as a power source, either through combustion or fuel cells.



E. Electric Vehicles (EVs):

Automobiles that use electrical energy as the primary power source.

battery / motor

F. Hybrid Vehicles:

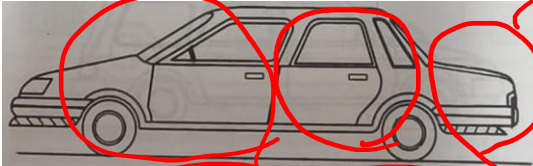
Vehicles that use two or more different power sources for propulsion.



battery + engine

Classification: Based on Body Style

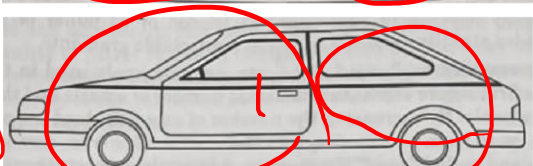
Saloon ✓
/Sedan ✓



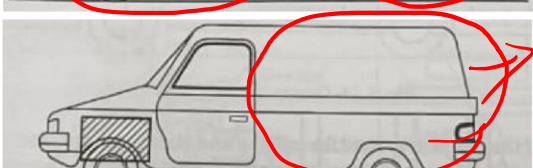
Hatchback
2-door ✓

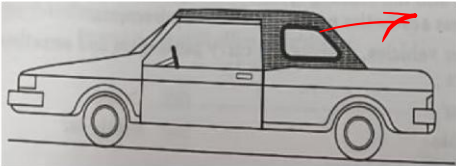


Coupe Car
Two-door

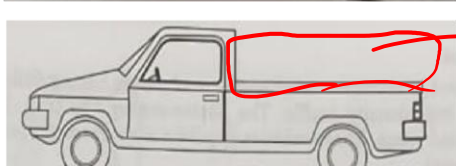


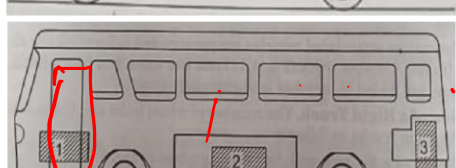
Van ✓











3 born deligh

Convertible ✓

Estate/wagon ✓

open Cargo

Pick-up ✓

Coach ✓



Manali

Scist

hatchback



Honda

city

Sedan

AEL ZC441 • Automotive Vehicles

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

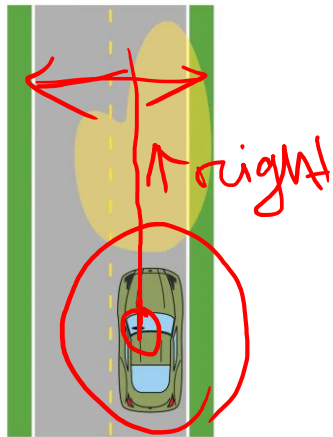
Classification: Based on Side of drive

innovate

achieve

lead

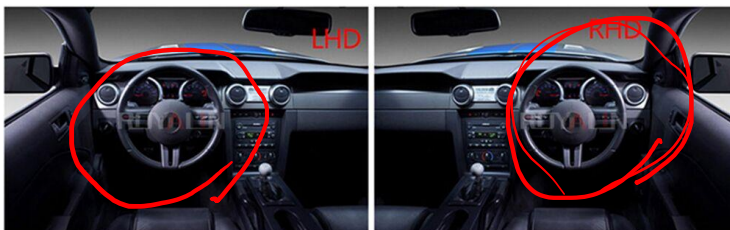
Left Hand Drive (LHD) ✓



Right Hand Drive (RHD) ✓



Why?



➤ LHD (Left-Hand Drive):

- Driver sits: left side ✓
- Vehicle: Right side of the road ✓

➤ RHD (Right-Hand Drive):

- Driver sits: Right side ✓
- Vehicle: Left side of the road ✓

Source: <https://www.worldstandards.eu/cars/list-of-left-driving-countries>



Classification: Based on transmission type



➤ Manual (Conventional) Transmission Vehicles:

- ✓ Gear ratios are changed manually by the driver.

Manual Car
AMT →

➤ Semi-Automatic Transmission Vehicles:

- ✓ Gear selection with partial automation.

FA →

➤ Automatic Transmission Vehicles:

- ✓ Gear ratios changed automatically during operation.

ABC
Ace Brake Clutch

Classification: Regulatory (Indian)



- ❖ Indian Classification: Governed by the Central Motor Vehicle Rules (CMVR) and the Automotive Industry Standards (AIS).

M = Passenger vehicles

N = Goods vehicles

L = Motorcycles / certain 3-wheelers & quadricycles

M1: ≤ 8 passenger seats + driver

M2: ≥ 9 passenger seats + driver; GVW ≤ 5 t

M3: ≥ 9 passenger seats + driver; GVW > 5 t

N1: GVW ≤ 3.5 t

N2: 3.5 t $<$ GVW < 12 t

N3: GVW > 12 t

Source: Official CMVR - MoRTH

Can't SUVs
small navigation
on seating
Annex
why?
Certification
prog.
Homologation

MORTH
CMVR
AIS
M $\rightarrow$ Passengers
N $\rightarrow$ Goods
L $\rightarrow$ 2 or 3-whe

BS $\rightarrow$ Broadcast stage

Session Roadmap

TAC → Type Approval Certificate
 prototype is approved
 COP → mass production
 ↓ type

2. Homologation ✓

Certification of a vehicle

OEM →
 TAC
 Hydrogen
 1 Prototype
 ↓

Petrol + Diesel vehicles
 Safety

What Homologation Means

❖ **Homologation:** Process of certifying a vehicle design before it enters production or sale.

➤ **Type Approval Certificate (TAC):** Prototype

➤ **Conformity of Production (CoP):** Mass-production

India: Primarily governed by the **Central Motor Vehicles Rules (CMVR)** along with applicable **Automotive Industry Standards (AIS)**.

HOMOLOGATION CERTIFICATION

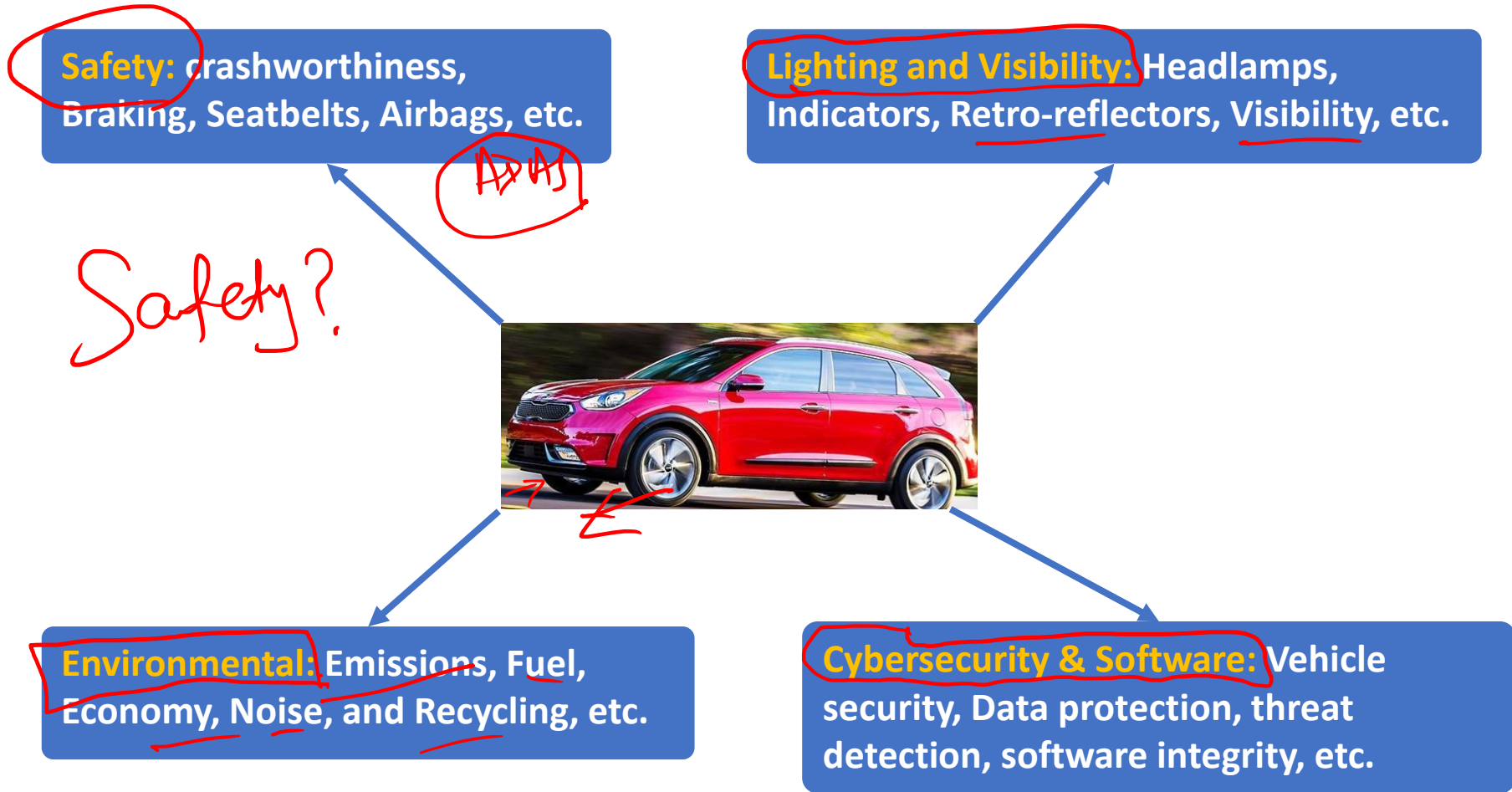


TAC

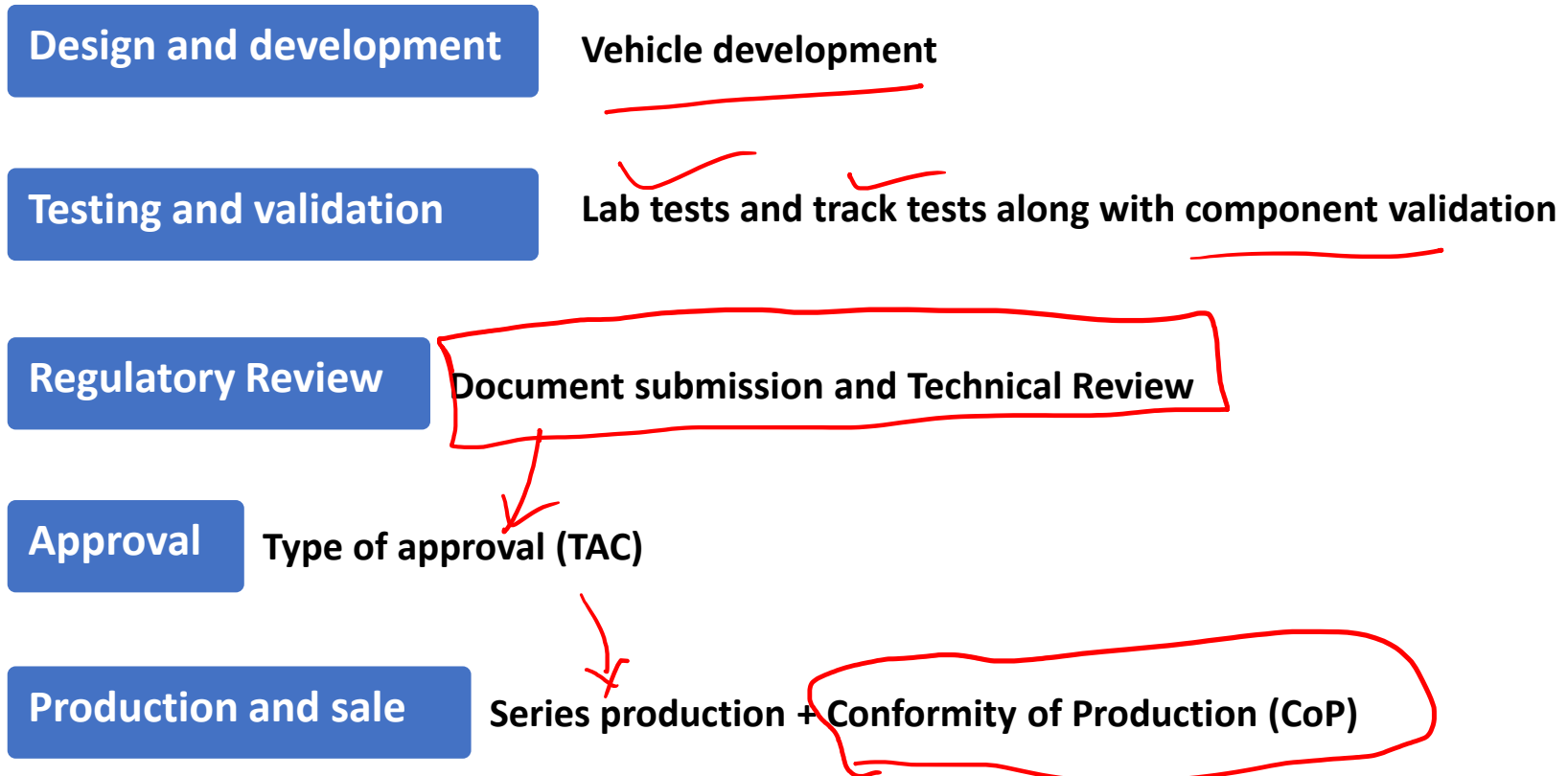
CoP →

Prototype → Regulatory Testing → Type of Approval → Production → Conformity of Production

Homologation: What is evaluated?



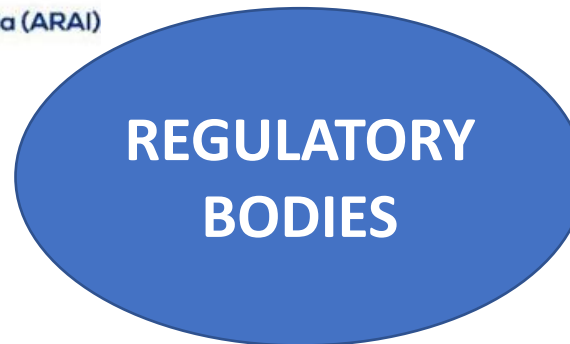
Homologation: Process



Vehicle Testing & Type-Approval Agencies



Mo RTH
AIS



Vehicle Testing & Type-Approval Agencies



Agency	Full Form	Location	Role
ARAI ✓	Automotive Research Association of India	Maharashtra ✓	Broadest all-round homologation + automotive R&D centre
ICAT ✓	International Centre for Automotive Technology	Haryana ✓	Comprehensive homologation + testing, especially strong in modern vehicle/electronic systems
GARC ✓	Global Automotive Research Centre	Tamil Nadu ✓	Comprehensive homologation with strong passive-safety, component and test-track capability
NATRAX ✓	National Automotive Test Tracks	Madhya Pradesh ✓	India's major proving-ground / vehicle-dynamics and track-testing centre
VRDE ✓	Vehicle Research & Development Establishment	Maharashtra ✓	Defence-vehicle R&D and evaluation centre - not primarily a commercial passenger-car homologation centre
CIRT ✓	Central Institute of Road Transport	Maharashtra ✓	Strongest emphasis on buses, commercial/public-transport vehicles and safety-critical components

Emissions Regulation



Emission norms specify how much pollutant a vehicle is allowed to emit

➤ Euro 6 / Euro 7 (Europe) ✓

Progressive tightening of NO_x, particulate, and now particle-number limits.

Euro 7 extends scope to brake and tyre particulates.

➤ BS-VI (India) ✓

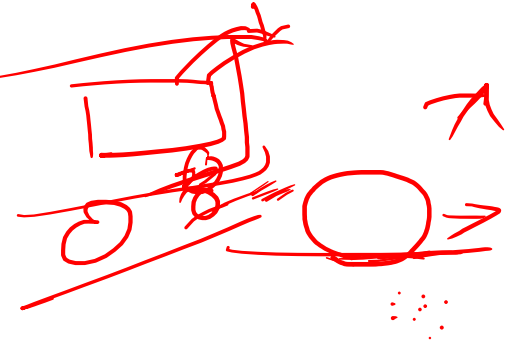
Aligned closely with Euro 6 stringency

➤ China 6

Combines elements of Euro 6 and US standards

➤ EPA (Environmental Protection Agency) Tier 3 (United States)

Regulates the vehicle emissions



How it is tested?

Drive cycles test: Comparisons

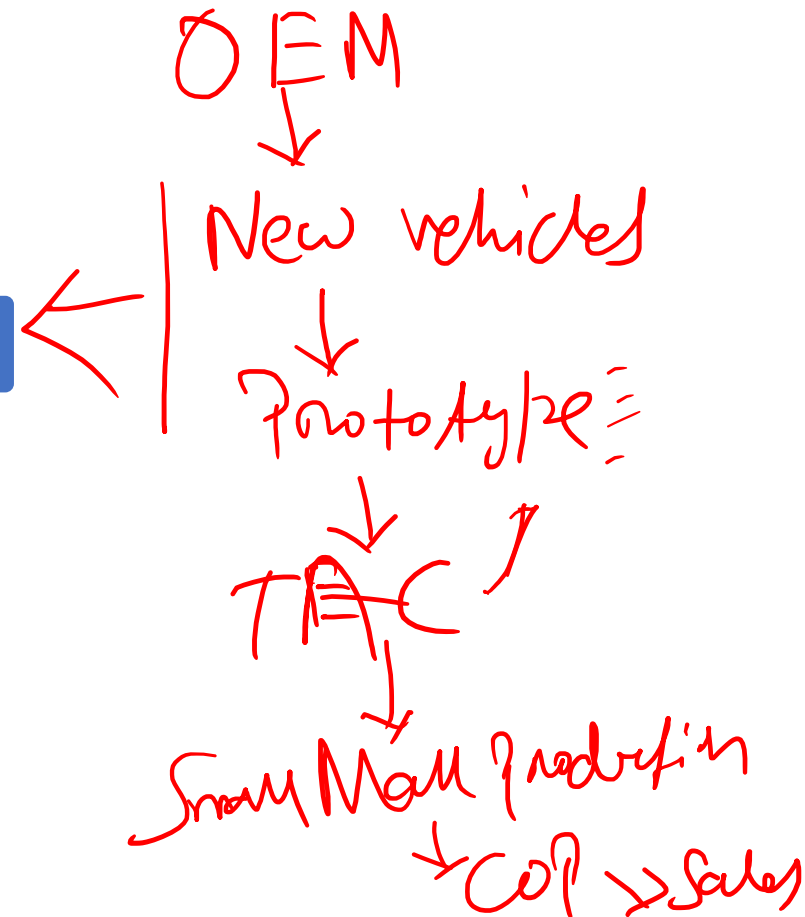
Drive cycle test: Variation of vehicle speed to test emissions and/or fuel consumption

Cycle	Region	Character	Key Limitation Addressed
NEDC ✓ (New European Driving Cycle)	Legacy EU ✓ (pre-2017)	Steady, gentle accelerations, low average speed	Widely criticized as unrepresentative of real driving
WLTP ✓ (Worldwide Harmonized Light Vehicles Test Procedure)	EU, India, others	Higher speeds, sharper acceleration, shorter idle <i>real condition</i>	Replaced NEDC to better reflect real-world driving
RDE (Real Driving Emissions)	EU, India (alongside WLTP) ✓	On-road testing with portable emissions equipment	Catches conditions no lab cycle can replicate
FTP ✓ (Federal Test Procedure)	United States	City driving cycle with a cold-start phase	Basis for EPA certification and fuel economy label

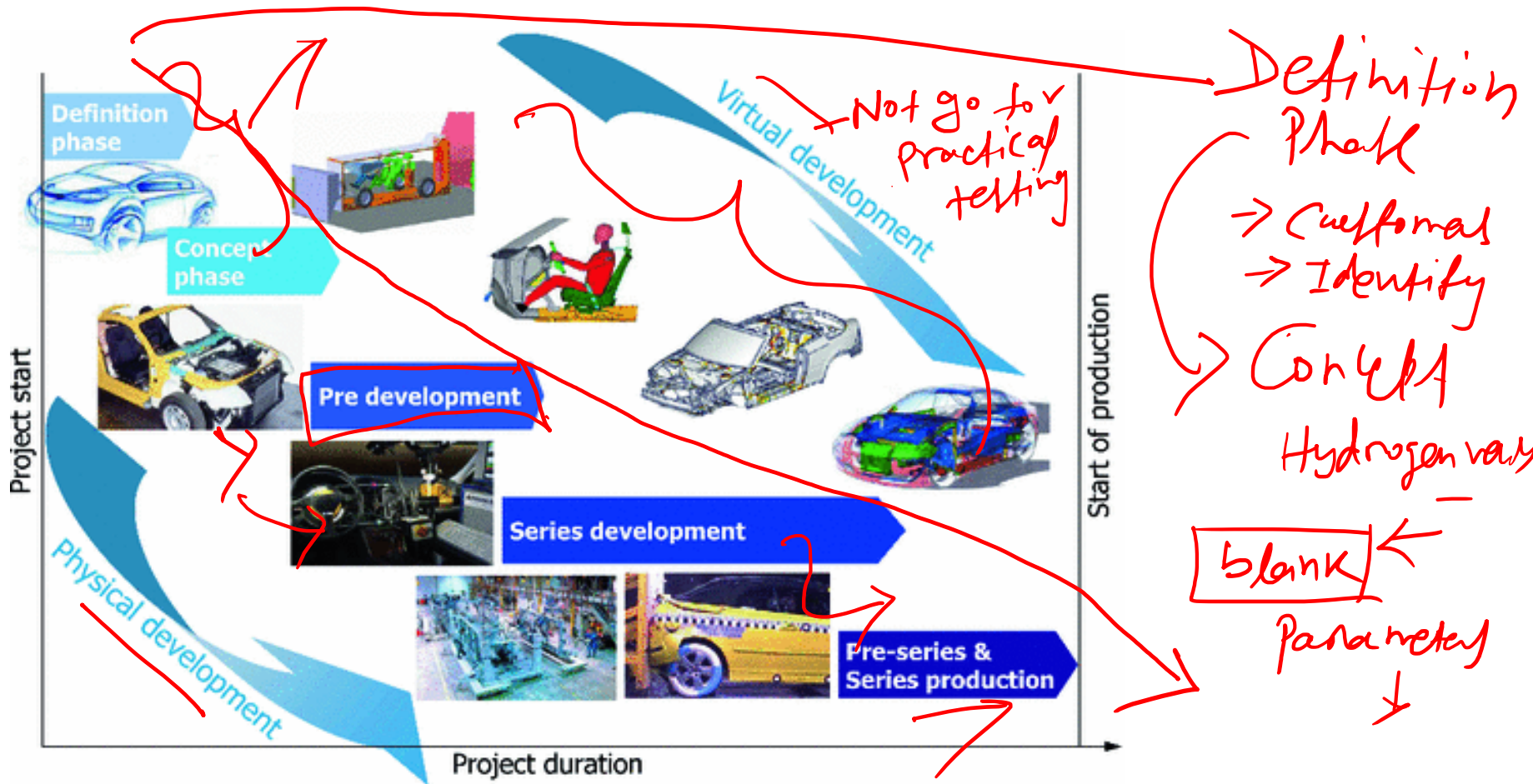
Handwritten notes:
 EQM T-3
 CARB → USA
 California Air Resources Board

Session Roadmap

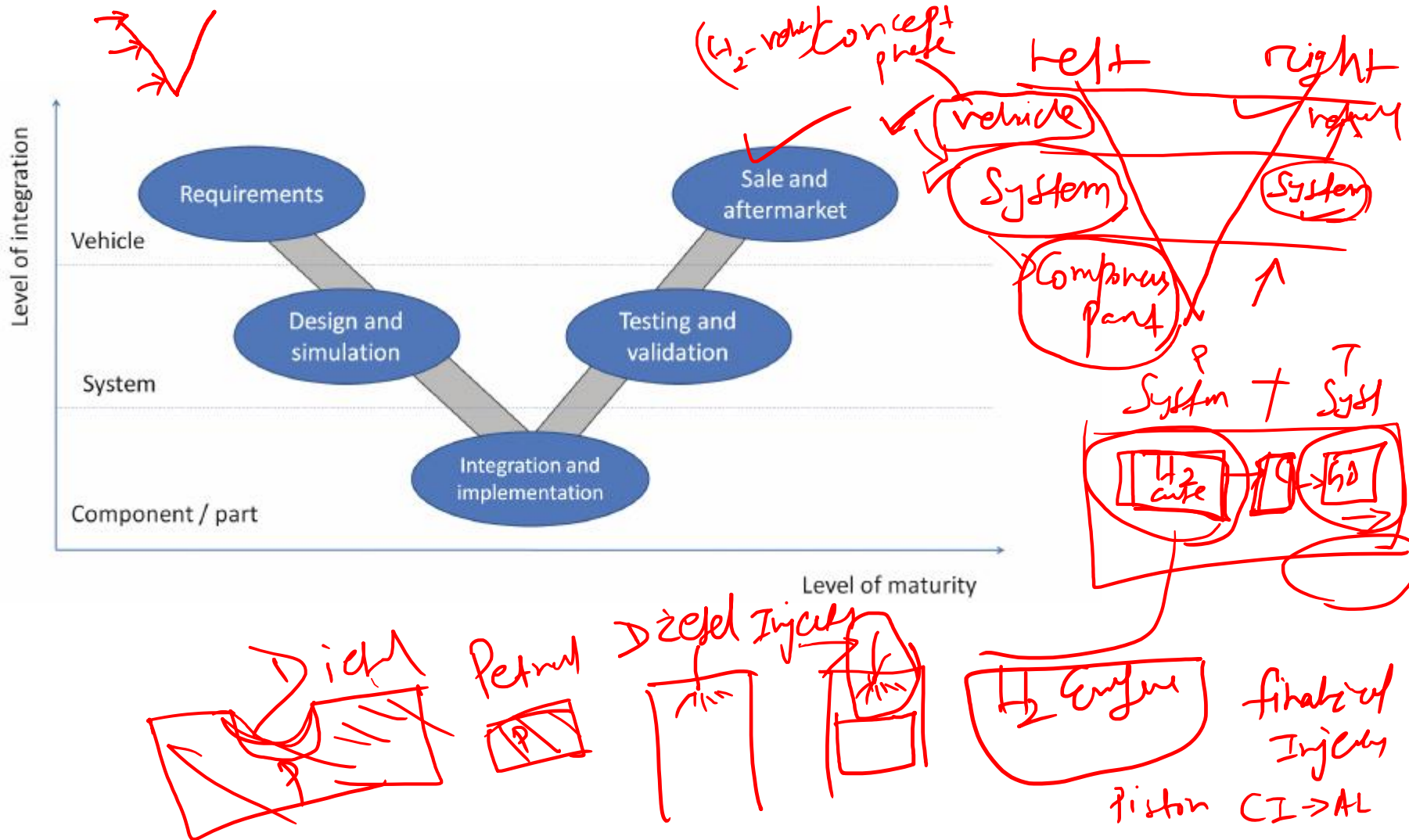
3. Development Methodology



Vehicle Development



Vehicle Development: The V-Model

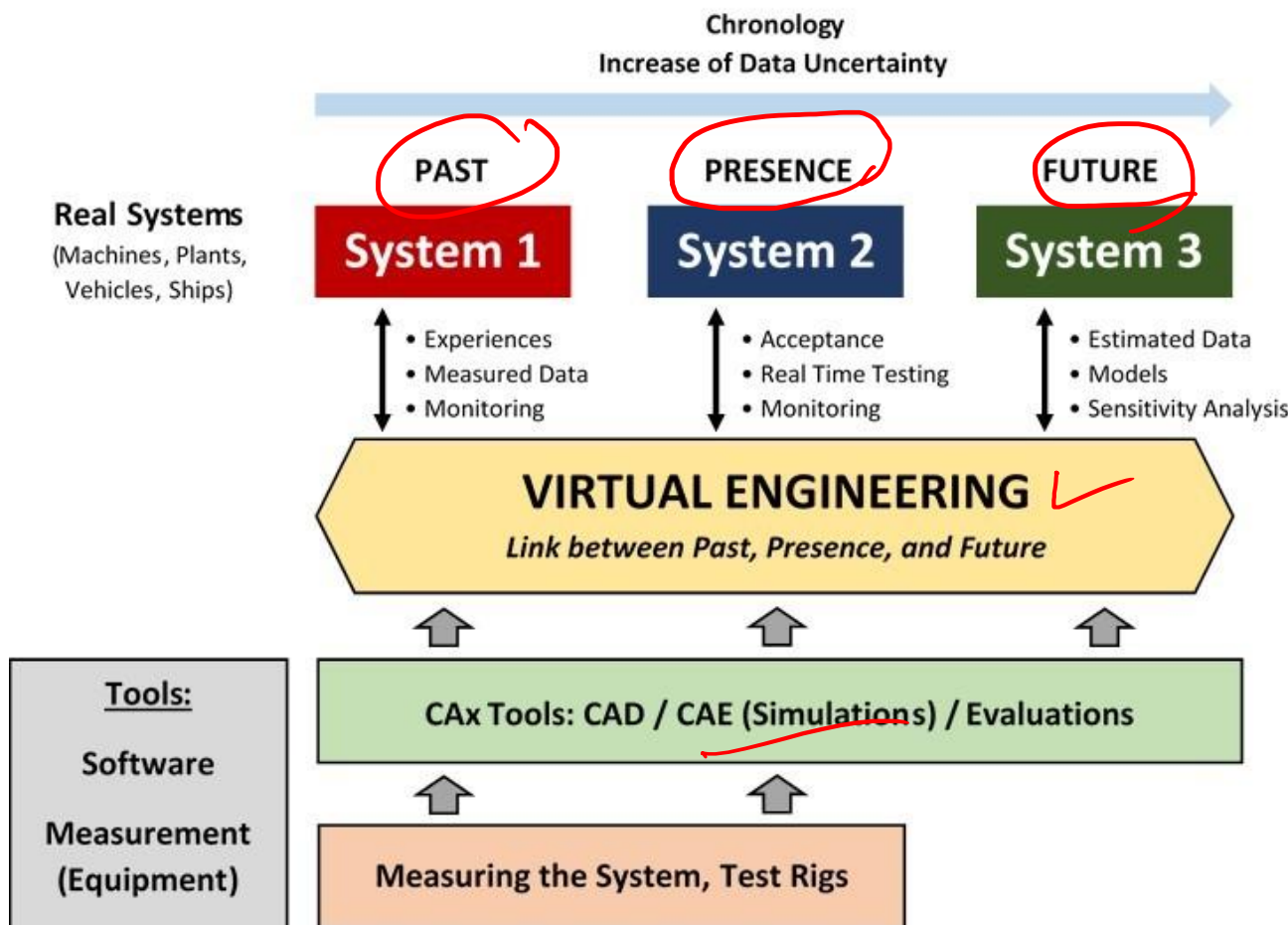


Session Roadmap



4. Virtual Engineering Methods

Why Virtual Engineering?



critical
→ Design parameters
→ Simulation

CAD

CFD

CAE

MIL/SIL/HIL

Simulation Disciplines

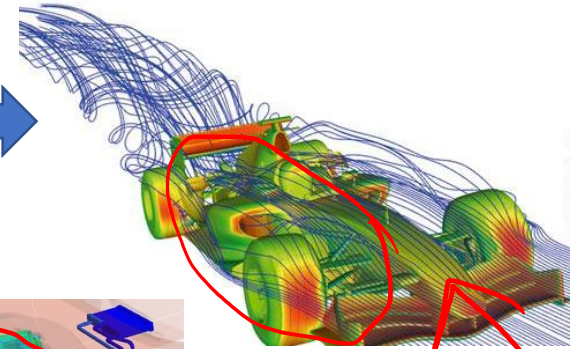
- **Fluid (CFD)** ✓
Computational Fluid Dynamics for aerodynamics and thermal management.
- **Structural (FEA)** ✓
Finite Element Analysis for crash, durability, and stiffness.
- **Controls / Powertrain (MIL/SIL/HIL)** ✓
Model-, Software-, and Hardware-in-the-Loop simulation for electronic control systems.

Aerodynamics and Thermal Management: CFD Simulation



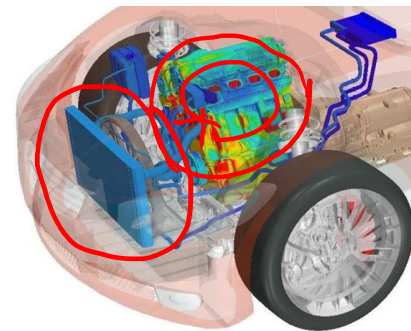
➤ External aerodynamics

Predicts drag, lift, and wake behavior



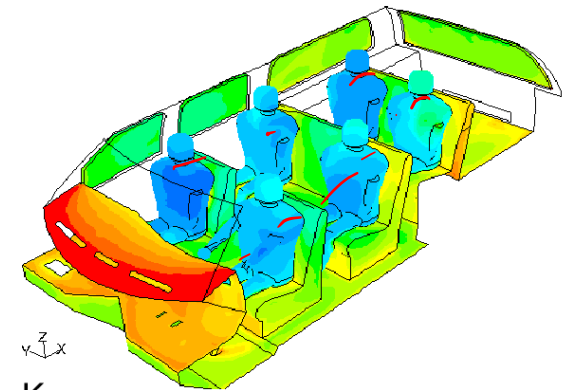
➤ Under hood and thermal management

Simulates airflow and heat rejection



➤ Cabin climate simulation

Predicts HVAC performance



Component level / System level / vehicle level

Structural Simulation (FEA)



➤ Crash simulation

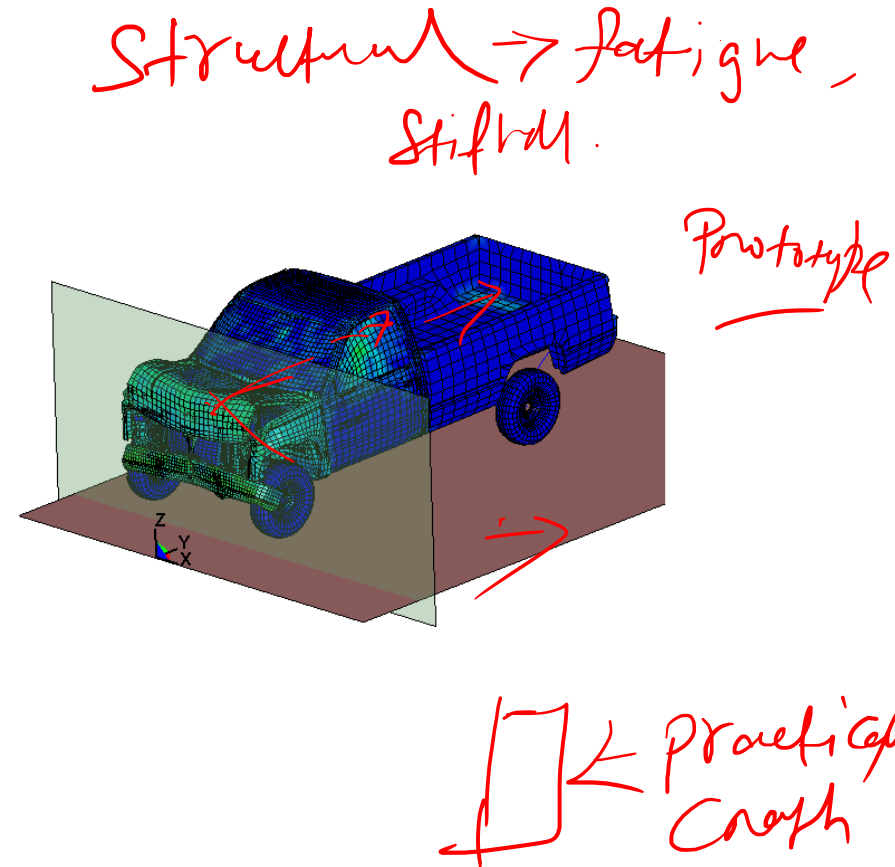
Explicit FEA solvers predict structural crush, intrusion, and occupant loading long before a physical crash test is booked.

➤ Durability / fatigue

Predicts component life under representative load histories, informing design before committing to tooling.

➤ NVH (Noise, Vibration, Harshness)

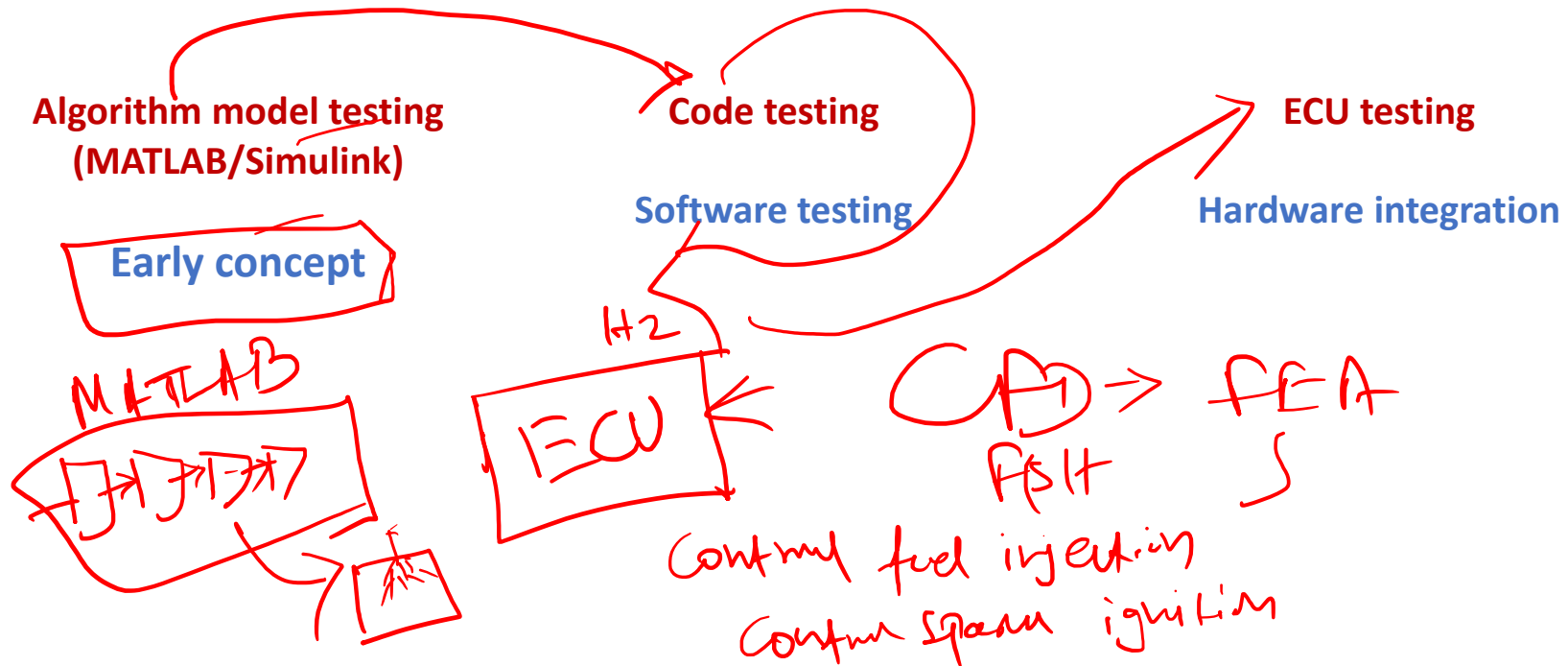
Modal and frequency-response analysis predicts resonances and structure-borne noise paths.



Controls / Powertrain (MIL/SIL/HIL)



From simulation to rear world testing



Vehicle Body and Chassis

*Skeleton → Chassis
Skin → Body*

Vehicle Body and Chassis



❖ Vehicle Structure: Chassis + Body

➤ Body

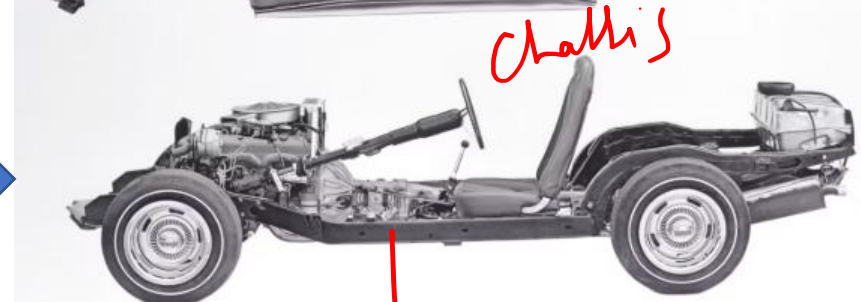
Structure enclosing the passenger/cargo space and providing shape, protection, aerodynamics and appearance.



body

➤ Chassis

Structural skeleton with major mechanical systems (Suspension, steering, brakes, powertrain, wheels, etc.).



chassis

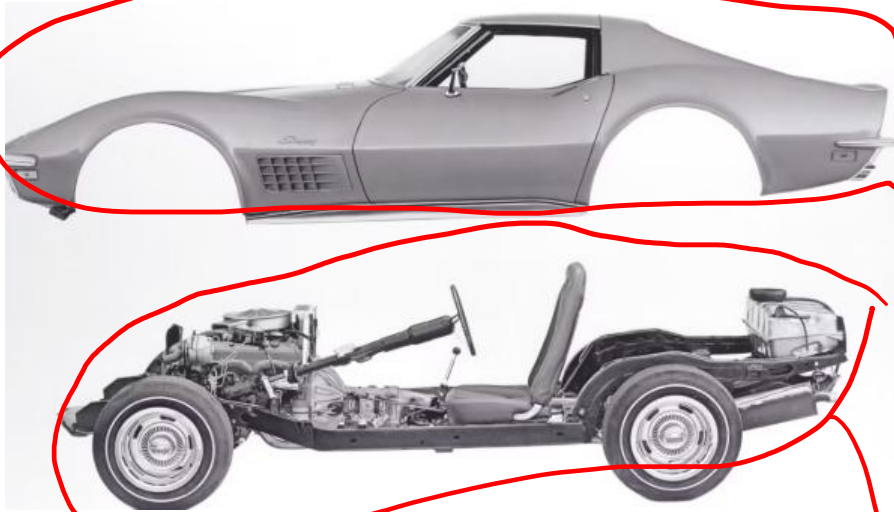
Chassis frame
+
V9 Components

Fundamental constructions



➤ Body-on-Frame Construction

Body and load-carrying frame are separate structures.



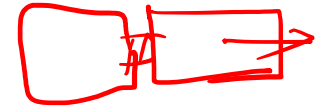
➤ Integral / Unibody Construction

Body and structural frame are integrated.



body-on-frame
or unibody

Body-on-Frame Vs Unibody



Parameters	Body-on-Frame Construction	Unibody Construction
Construction	Body mounted on a structural frame	Body and chassis form a single integrated
Load carrying	By structural frame	By both <u>Body</u> and <u>chassis</u>
Weight	Higher	Lower
Fuel Efficiency	Lower	Better
Towing / Payload	Excellent	lower for comparable vehicle classes
Repair	Easily repaired/replaced	Expensive to repair
Applications	Trucks, pickups, and off-road vehicles	Passenger cars, crossovers, many SUVs

- **Body-on-Frame: Strength + Payload/Towing + Durability**
- **Unibody: Lightweight + Handling + Efficiency**

Unibody construction

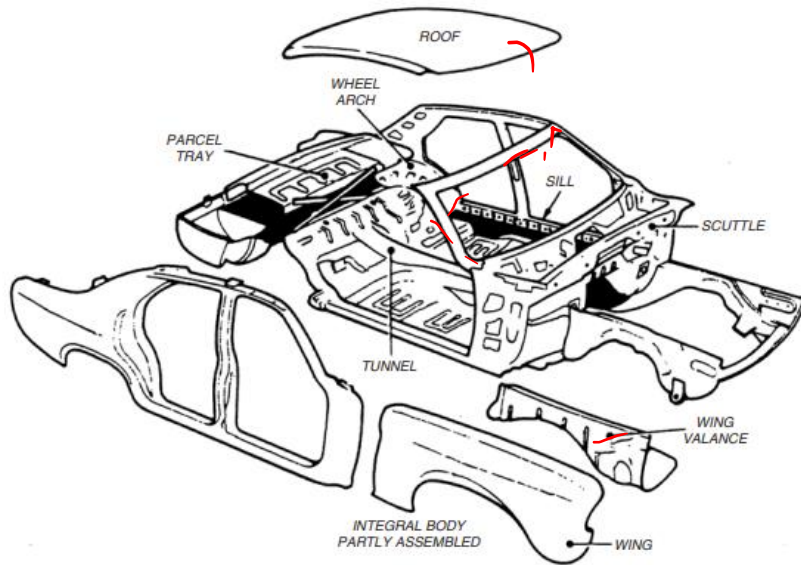


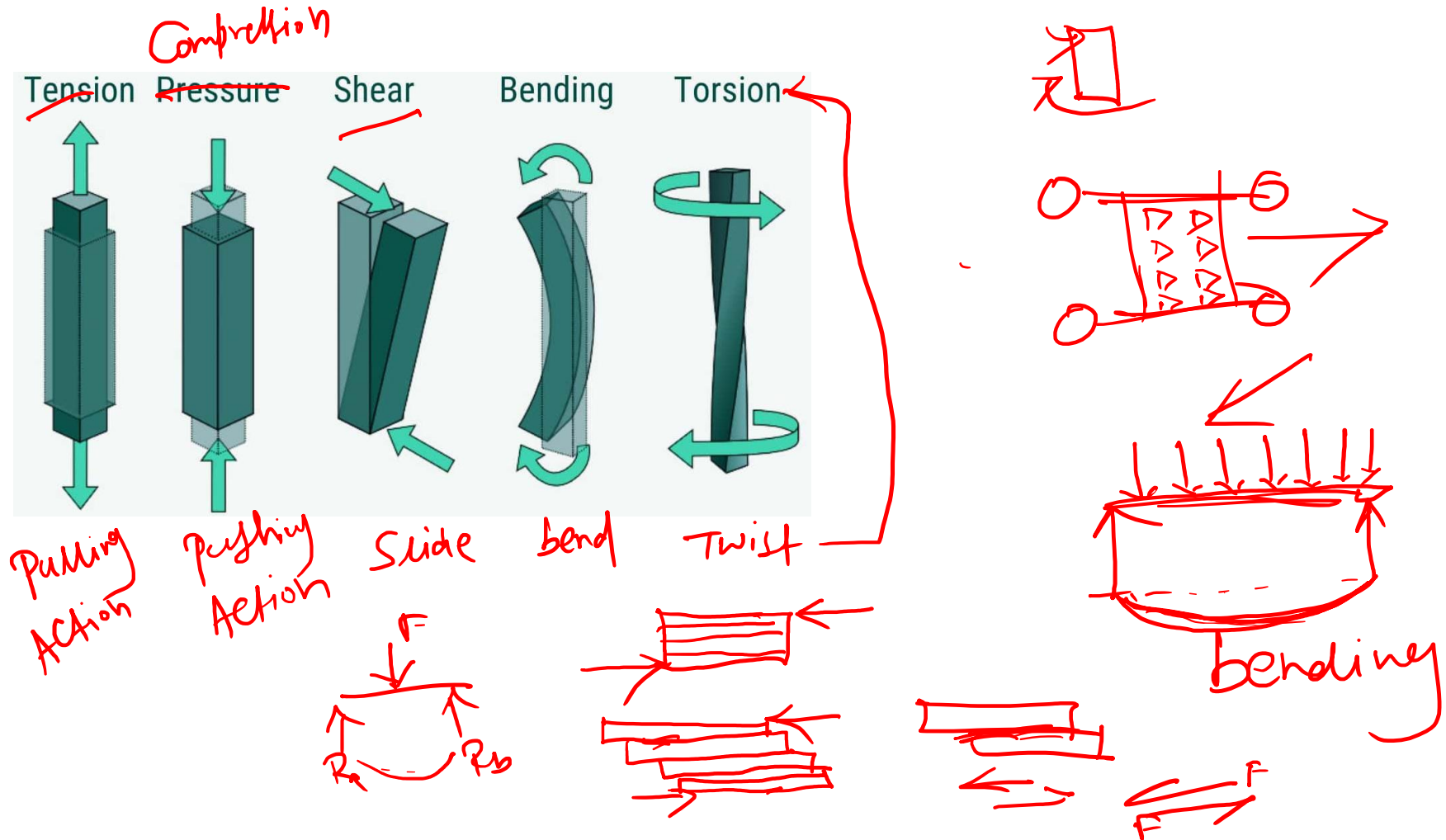
Figure: Frameless/Integral
body construction

Table: Thickness of different sections

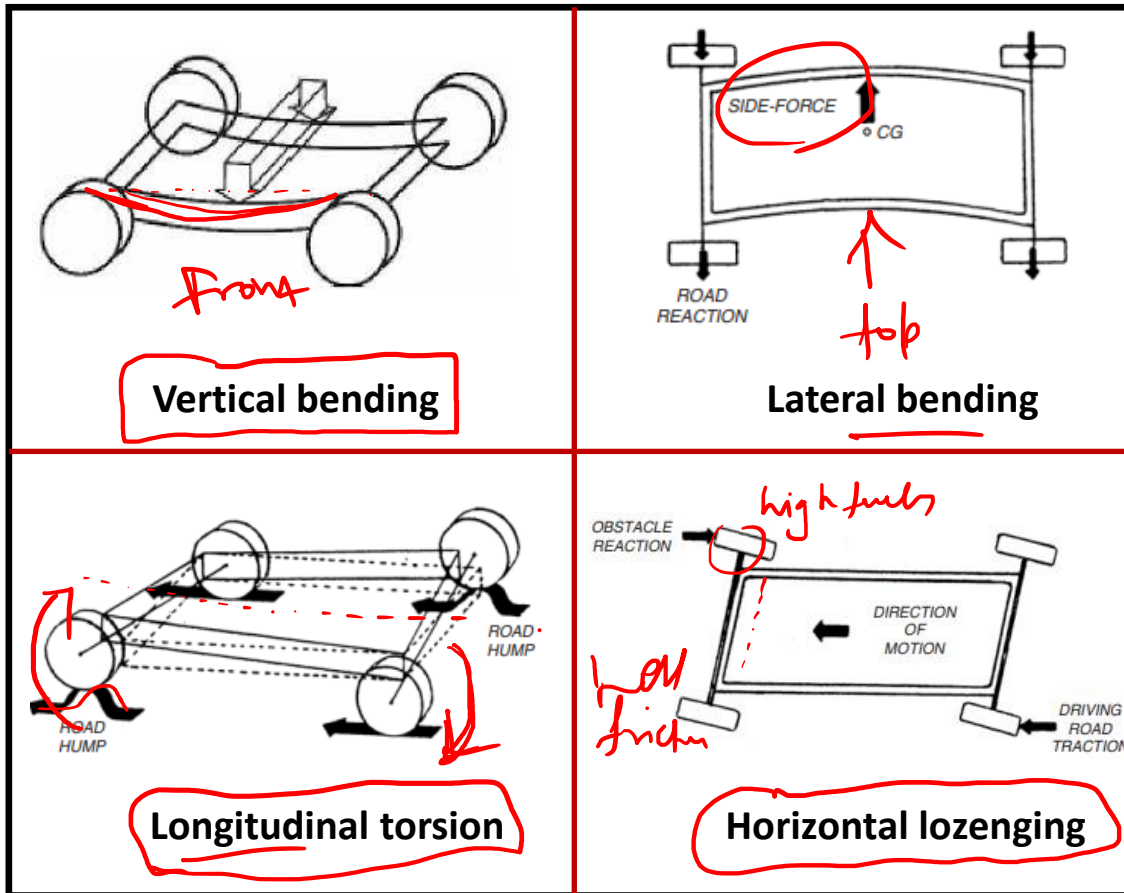
Component / Application	Typical Thickness (mm)	Remarks
Large-area panels (<u>roof</u> , <u>quarters</u> , <u>doors</u> , and <u>floors</u>)	0.915	Subjected mainly to <u>direct strain</u>
Panels with <u>curved contours</u>	0.765 – 0.66	Curvature improves stiffness and rigidity
Major <u>structural parts</u> (<u>cross members</u> , <u>rails</u> , and <u>pillars</u>)	1.00 – 1.25	Load-carrying structural members
<u>Local reinforcements</u>	1.625	Used where <u>additional strength</u> is required

0.6 → 1.6 mm

Loading on Chassis



Loading on Chassis



- Vertical bending:**
Vehicle weight
- Lateral bending:**
Cornering forces
- Longitudinal torsion:**
Different road heights
- Horizontal lozenging:**
Unequal forces



Vertical bending of Chassis



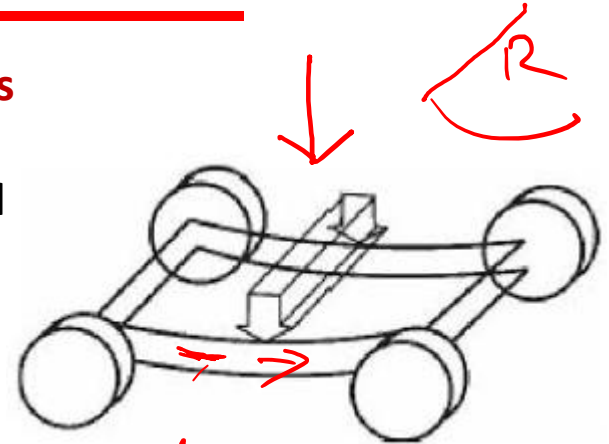
❖ Vehicle components and payload apply downward forces

➤ The central portion of the chassis tends to **sag downward**

Bending equation

$$\frac{\sigma_b}{y} = \frac{M}{I} = \frac{E}{R}$$

✓ $E \rightarrow$ still
 $I \rightarrow$ effectively used



Where:

σ_b = Bending stress: Internal stress developed in the material to resist bending

M = Bending moment: How strongly the load tries to bend the chassis

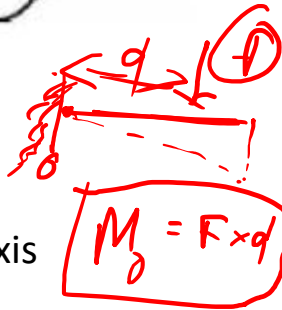
I = Moment of inertia: How effectively material is distributed away from the neutral axis

y = Distance from neutral axis: How far a material point is from the neutral axis

$Z = I/y$ = Section modulus: Ability of cross-section to resist bending stress

E = Material resistance to elastic deformation = How stiff the material

R = Radius of curvature after bending



Bending stiffness (EI): How difficult to bend the member

Design focus: Higher section modulus, Lower bending stress, higher bending stiffness

